

**How Does AI's Automation Technology Affect Employment
In the Software Engineering Field?**

EST 441

Final Paper

Donghun Yoo

114000660

1. Introduction

When I returned to campus for the fall semester of 2024 following military service, the changes I observed were immediate and striking. All students were using AI tools for their daily activities of writing, coding, and conducting research. This was not an experimental effort by early adopters, but instead represented a quietly normalized mode of operation. These experiences prompted more specific questions about the effects of AI on the job market for which I was preparing to compete. During this semester, I will receive a degree in technological systems management with a specialization in computer science and compete for a substantially different market for software engineering than existed 2 or 3 years ago. Entry-level recruitment by major technology firms has declined substantially, and tasks that previously characterized work by junior developers are now largely performed by AI tools. Understanding the implications of these changes is not an abstract exercise, but a practical one. This analysis of data describing patterns of labor market outcomes, productivity, and occupational experience for the period 2023 to 2026 supports the view that the effects of AI automation on employment in software engineering are not uniform across sectors. Instead, the degree of experience, expertise, and type of work performed by individual developers strongly influences the capacity of AI tools to replace existing tasks of employment.

Work by junior developers is declining rapidly as AI tools replace tasks previously requiring such workers. The productivity of highly experienced engineers is increasing substantially, and as a result, they provide significantly greater value to their employers. The emergence of entirely new categories of AI professional experience further distinguishes this market from the general field of software engineering. Finally, this paper describes the magnitude of the resulting differences in wages and considers the implications of these patterns for future directions of these industries.

Section 2 describes the background of AI technology and its development. Section 3 analyzes the differences in employment impact by level of experience and role type. Section 4 examines the broad structural implications for the industry. Section 5 addresses the limitations and counterarguments of this analysis before concluding.

2. The Evolution of AI Technology

2.1 Defining AI and Large Language Models

Artificial intelligence encompasses computer systems that perform tasks requiring human intelligence, such as pattern recognition, natural language generation, decision making, and reasoning (Google Cloud, n.d.). The technology most directly applicable to software engineering work is the large language model. These systems are trained on large volumes of text and code and generate outputs that closely approximate human-written text. They support tools such as GitHub Copilot for real-time assistance with code development and ChatGPT for the generation of responses appropriate to the content and context of natural language questions. Awareness of the principles underlying these systems is essential for assessment of their effects on skilled performance of knowledge work that is largely based on pattern recognition, contextual judgment, and fluency of natural language.

The generality of large language models represents a major departure from previous systems of artificial intelligence, which were generally narrowly adapted to particular applications such as playing chess or filtering e-mail. In contrast, systems trained on large and diverse volumes of text and code assist with tasks of writing, summarizing, translating, and developing code without additional adaptation to particular applications. As a result, they are applicable throughout the entire workflow of software development and affect not only individual tasks, but also many of the activities characteristic of the daily work of a junior developer.

2.2 Historical Development of AI

The trajectory to current capabilities of AI was not abrupt, but based on ideas of neural networks, machine learning, and pattern recognition developed since the 1950s and limited for many decades by available computing power and data (World Economic Forum, n.d.). Important milestones were achieved in the 1990s with the defeat of a world champion at the most complex game of strategic interaction ever devised. That achievement represented the first demonstration that machines could exceed human performance in such a game. However, the resulting system applied only to the game of chess. A greater transformation occurred in the 2010s, when progress in deep learning and the availability of large data sets enabled the development of systems applicable to multiple domains. Defeat of a world champion at the game of Go confirmed the capacity of AI to manage levels of strategic complexity previously considered inaccessible to automated methods. Subsequent availability of AI-based tools for everyday use provided access to individuals with no technical training and initiated a new phase of interaction between AI and professional work.

Use of a framework for understanding these changes illustrates the validity of Bell's law, which states that major new modes of computing emerge approximately every 10 years and are more widely applicable and integrated into everyday life than previous modes. Use of large computer systems replaced use of smaller ones, which in turn replaced use of personal computers, followed by the development of mobile phones and other devices. Each stage of development produced new categories of work and disrupted previous categories. The present development of AI is consistent with this overall pattern of change. However, its capacity for disruption extends to cognitive and professional domains that remained largely unaffected by previous applications of automation.

2.3 Research Projections and Labor Market Implications

A 2024 survey of about 2,800 AI researchers revealed that many believe that AI will achieve human-level performance on most cognitive tasks within a few decades, and possibly

within 10 years (Grace et al., 2024). These projections reflect not the certainty of particular timelines, but rather the proximity to actual work on AI development. As a result, expectations of continued rapid increases in capability are strongly influencing corporate plans for investment, staffing, and operations, regardless of whether the most optimistic projections ultimately prove valid.

More conservative estimates of short term progress are provided by analysis of occupational exposure to large language models (Eloundou et al., 2023). The results confirm that software engineering represents an occupation of very high exposure, in which a large fraction of core tasks potentially could be carried out or supported by large language models. However, the authors emphasize that exposure does not imply full automation, and that occupational effects of large language models represent an order of magnitude greater than most prior expectations. Finally, these results illustrate the value of shifting the discussion away from speculation about long term displacement toward empirically based evaluation of the extent to which various tasks can be achieved with current technologies.

3. Divergent Impacts on Software Engineering Employment

3.1 Entry-Level Developers

The most striking evidence that artificial intelligence is transforming employment in software engineering is at the entry level. Rates of employment for developers aged 22 to 25 declined by almost 20% from their 2022 peak (Stack Overflow, 2025). Junior hiring fell by 25% for the largest 15 technology firms between 2023 and 2024 (Stack Overflow, 2025). These results reflect an actual contraction in a category of positions that previously had served as the standard starting point for computer science graduates. The underlying mechanism of this contraction is straightforward.

AI-based tools such as GitHub Copilot, Amazon CodeWhisperer, and similar systems automate large components of work previously performed by junior developers, including completion of code, generation of boilerplate, development of tests, and implementation of well specified features. A survey of hiring managers showed that 70% believed that AI tools already were capable of replacing work that otherwise would have been performed by an intern or entry-level developer (Stack Overflow, 2025). The degree of accuracy of that perception was less important than the effect on actual patterns of hiring. Organizations that previously would have employed three junior developers now generally employ one, and use AI tools combined with close supervision to compensate for the difference. This illustrates a broader concern of potential importance beyond immediate estimates of employment.

The experience of training junior developers has previously provided a basis for the development of more advanced skills required by the field. Reductions in entry-level hiring may ultimately lead to disruptions in the supply of experienced engineers. Current estimates of these effects are incomplete, in part because of the long time periods required for their expression. However, Autor et al. document similar patterns of disruption of the supply of mid level skills associated with prior episodes of technological change (Autor et al., 2003). Continued application of the present pattern of technological substitution would ultimately produce an analogous effect on the structure of software engineering employment.

3.2 Senior and Experienced Developers

Employment of developers over age 30 increased 6% to 13% over the same period that hiring of junior developers declined (ADP Research Institute, 2025). This divergence reflects major differences in the ways experienced engineers interact with AI tools. Rather than replacing work performed by senior developers, AI increases its efficiency. Experienced engineers provide judgment, knowledge of system architecture, and the ability to interpret ambiguous requirements that are not reliably matched by current AI technologies. As a result,

experienced developers achieve substantial increases in output and productivity without increasing headcount.

My interpretation is supported by controlled studies. Developers who used GitHub Copilot completed assigned programming tasks at a rate 55.8% faster than did developers without AI assistance (Peng et al., 2023). Implementation of AI tools in existing work processes also increased the number of completed tasks per engineer by 26% (Peng et al., 2023). Estimated reductions in time required for software development ranged from 20% to 45% for organizations as a whole (McKinsey & Company, 2023), and were greatest for experienced developers. These increases in productivity reduced the need for increases in headcount.

In effect, the output achieved by one highly experienced developer with AI assistance was equal to that achieved previously by a larger group of less experienced developers. This pattern explains the continued expansion of employment for experienced developers and the reduction in hiring of junior developers. The same level of organizational output can be achieved with a smaller group of highly experienced developers and with greater use of AI technology. This results in increased security of employment for experienced developers but eliminates conventional opportunities for entry into the field.

3.3 AI Specialized Engineers

The rapid expansion of AI adoption has created an entirely new set of engineering positions. Machine learning engineers, AI systems developers, and specialists in AI integration have evolved from a niche to among the most aggressively recruited technical positions across all industries. Demand for machine learning engineers increased by 45% annually through 2025, and the number of job openings requiring knowledge of AI or machine learning increased 150% between June 2024 and June 2025 (Stanford Human-Centered Artificial Intelligence [HAI], 2026). These positions are now available not

only for firms specializing in AI, but also for companies in health care, logistics, financial services, and retail, as each sector attempts to incorporate AI capabilities into existing operations. Salaries associated with these positions reflect the degree of demand.

Average salaries for AI engineers reached \$206,000 in 2025, representing an increase of approximately \$50,000 over the previous year (Stanford HAI, 2026). Nearly 90% of firms surveyed developed new positions within their engineering departments devoted exclusively to AI (Stanford HAI, 2026). These positions required integration of expertise in machine learning methods, experience with procedures for training and evaluating models, and the capability for designing systems that incorporate outputs of AI applications into overall technical configurations. They were accessible to individuals with strong backgrounds in fundamental computer science who were willing to develop high levels of specialization, but were not readily achievable. Rates of progress were so rapid that knowledge of all aspects of these fields became obsolete within 1 to 2 years.

The resulting rates of growth for AI related occupations were sufficient to support predictions that these positions would represent among the most rapidly growing occupational categories worldwide for the 20 years through 2030 (Future of Jobs Report 2025, 2025). Net rates of employment growth would remain positive even in the face of competitive pressures for the displacement of other occupational categories. Projected growth in employment for software developers as a whole was also highly favorable, and predicted an overall rate of employment growth of about 25% for the 12 years through 2032 (U.S. Bureau of Labor Statistics, 2025). However, these results reflect conditions that existed before the full incorporation of the current phase of AI driven restructuring into records of employment conditions. The resulting contrast between expectations of continued high rates of growth and observations of substantial reductions in employment for junior technical positions provides

additional evidence of the extreme rapidity with which underlying conditions for the field continue to change.

3.4 Wage Polarization and Structural Inequality

The data on employment for these three groups reflect one aspect of the ongoing transformation. Data on wages reflect another. General software engineer salaries have remained relatively stable over the past 2 years in the face of documented gains in productivity attributable to AI tools. In contrast, salaries for specialists in AI have risen substantially, with senior AI engineers at large technology firms earning about 70% more than general software engineers with comparable levels of experience (ADP Research Institute, 2025). This reflects a relatively recent trend that is likely to continue.

The persistence of low rates of wage adjustment in the face of large productivity gains illustrates a pattern previously observed for previous waves of automation (Autor et al., 2003). Increases in worker productivity associated with new tools generally result in large short term gains for employers that are not reflected in increases in wages. Thus, a developer who produces 55% more code per day with the assistance of Copilot does not receive a 55% increase in salary, but conveys substantial additional benefits to the firm. Use of AI tools is now essential for competitive survival in the job market, but provides little or no benefit to individual earnings.

Finally, Eloundou et al. find that occupations most affected by large language models generally are high wage, high skill positions (Eloundou et al., 2023), in contrast to previous experience with automation that primarily affected low wage, routine based occupations. As a result, the effects of AI on wages in the field are likely to be concentrated primarily at mid and high wage levels, rather than at low wage levels as previously experienced with automation. This constitutes an important departure from earlier experiences of labor market disruption and warrants additional attention as data on long term trends accumulate.

4. Broader Structural Implications

4.1 Industry-Wide Projections

Although segment-level trends may appear to reflect volatility, aggregate data on software engineering employment convey a substantially more stable picture. Increases in software related and AI specialized jobs, however, do not necessarily result in growth for the conventional trajectory from junior to senior development careers, and may obscure associated patterns of compositional change. Continued growth of employment for software developers is projected through 2032, driven by demand concentrated in health care, finance, and infrastructure sectors (U.S. Bureau of Labor Statistics, 2025). Globally, rates of growth for AI and machine learning jobs are similarly among the highest for any occupational category (Future of Jobs Report 2025, 2025). But the changes in composition associated with these overall increases in employment are not captured by the traditional classification of jobs into categories of skill, education, and training.

Rather, McKinsey & Company describes AI's effects on work as changes in patterns of task exposure (McKinsey & Company, 2023). Most occupations (including software engineering) will experience some degree of automation of tasks, but will not be completely displaced. Over time, this results in a redefinition of job roles as a practical consequence: Different tasks become incorporated into the same job title, and are associated with higher requirements for entry-level qualifications. This pattern is already evident in job advertisements for software engineering positions, where expertise with AI tools is routinely expected and not considered to be a competitive advantage.

4.2 Skill Gap and Educational Responses

One consequence of the current transition is an increasing disparity between what most computer science programs teach and what the market rewards. Traditional curricula

covering data structures, algorithms, systems programming, and principles of software engineering remain essential, but do not adequately reflect the skills distinguishing AI specialists from general developers. These include methods for evaluating models, engineering prompts, designing data pipelines, and integrating AI-based systems. The expertise required to achieve high performance in the rapidly growing, highly remunerative segments of the market is not yet incorporated systematically into undergraduate programs.

As a result, the lag between market requirements and educational responses is not unusual and has occurred with each prior transition to new technologies. However, the rate at which the gap between market requirements and program outputs is developing is unusually rapid. The disparity between what the market rewards and what programs produce developed within 2 to 3 years, rather than over the longer periods characteristic of previous transitions. Programs are being revised rapidly, but the time required to develop qualified faculty and validated course materials remains substantially greater than the rate of market change. Consequently, individuals who pursue intensive, self-directed education in methods of AI system development and in principles of machine learning are at a substantial advantage relative to individuals trained solely by traditional programs.

5. Limitations and Counterarguments

Important limitations apply to this analysis. Most of the employment and wage data presented here reflect U.S. experience and may differ substantially from global trends. In settings of low labor costs, AI technologies are likely to augment rather than replace human developers and generate different patterns of employment outcomes. The limited information on the effects of AI on software engineering labor markets for different national conditions limits the generalizability of these results.

The patterns identified by the secondary data analysis approach reflect aggregate, but not complete, explanations of firm or individual level decisions. Contraction of entry-level hiring is largely attributable to overall reductions in costs for the technology sector that occurred following the large expansion of hiring in 2021 and 2022. Thus, it reflects largely concurrent, rather than purely causal, effects of AI adoption. Separation of causal effects attributable to AI from simultaneous influences of macroeconomic conditions remains methodologically difficult with available data. As a consequence, the claims of causality are based primarily on patterns of association and plausible mechanisms, rather than on results of controlled experiments.

Finally, this section considers alternative interpretations of these results that challenge the validity of the displacement narrative. Some economists who study automation contend that initial disruptions of employment associated with new technologies ultimately generate substantial increases in demand for new types of work. The history of computing illustrates this principle: Elimination of clerical positions attributable to automation of manual calculations ultimately resulted in substantially greater demand for work in software development, data analysis, and systems engineering. Whether the application of AI will also generate patterns of increased demand for previously non-existent types of work remains to be determined. The present inability to resolve this issue reflects the fact that application of AI constitutes a fundamentally different type of technological change than has occurred previously with automation of physical or routine clerical tasks (Eloundou et al., 2023).

Additional information supporting these conclusions will require continued follow up over substantially longer periods of time. The rapid pace of the present transition will also make it difficult to distinguish between the effects of cyclical and structural change. Reductions in entry-level employment constitute the first category of labor affected by

economic contractions and the last to be restored during periods of economic recovery. The present period of decline coincides with broader contractions of the technology sector and therefore limits opportunities for clean attribution. Subsequent work based on firm-level data, longer time series, and cross national comparisons would greatly increase the accuracy of these conclusions and permit a more direct assessment of the independent effects of AI on patterns of employment.

6. Conclusion

The available evidence does not support a single narrative of the impact of AI on software engineering employment. Entry-level positions are contracting and reflect the increasing absorption by AI tools of tasks previously requiring junior developers. Senior developers are more productive and currently less likely to be unemployed. Specialists in AI represent a separate level of the market with patterns of demand and compensation consistent with the growing importance of AI capabilities for virtually all sectors. Disparities in wages reflect and extend beyond the limits of aggregate measures of employment.

For those entering rather than observing the field, the overall message is that the value of a generalist education in computer science remains intact. However, the requirements for what must be built upon that foundation have changed. The resulting patterns of specialization represent a major source of competitive differentiation in the current market. A strong foundation of education in computer science remains essential, but subsequent patterns of performance depend increasingly on additional training in methods of development for AI-based systems.

Major limitations of this analysis should be emphasized. The results are largely based on U.S. experience. The field is evolving at a rate substantially greater than the capacity of most cycles of research to capture. The ability to isolate mechanisms of cause and effect was

further limited by the retrospective nature of secondary data analysis. Nevertheless, the data support a major descriptive conclusion: Effects of AI on software engineering differ according to levels of experience, patterns of specialization, and types of professional roles. These differences are substantial, well reflected in available data, and likely to increase further as the capabilities of AI continue to develop. Finally, the ultimate positions within the overall pattern of disparity depend strongly on current decisions regarding the allocation of resources to further development of technical capabilities.

7. AI Statement

I used generative AI tools at multiple stages of conducting and writing this paper. AI primarily aided with organizing section structures and generating initial versions of literature review paragraphs. For the section on wage polarization, the AI-generated draft emphasized the benefits of specialist AI wages and portrayed the current situation as an opportunity for career advancement. I subsequently revised that section to reflect continuing wage stagnation for general software engineers and to illustrate the failure of productivity gains from AI to translate into higher wages for workers using those tools. This provided a more complete and empirically accurate description of the results. In addition, I largely prepared the discussion of broader structural implications and of study limitations, because the quality of AI-generated content for these sections was substantially less than for other parts of the paper. I organized and verified all aspects of argument structure, selection of sources, and in-text citations, and performed final revisions to achieve consistency with the results of the studies and to eliminate modes of presentation that conveyed an impression of advocacy rather than of analytical interpretation.

References

- ADP Research Institute. "Workforce Vitality Report: Technology Sector Employment Trends." ADP, 2025.
- Autor, D. H., Levy, F., & Murnane, R. J. (2003). The Skill Content of Recent Technological Change: An Empirical Exploration. *The Quarterly Journal of Economics*, 118(4), 1279–1333. <https://doi.org/10.1162/003355303322552801>
- Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2023, March 17). *GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models*. arXiv.org. <https://arxiv.org/abs/2303.10130>
- Google Cloud. "What Is Artificial Intelligence (AI)?" Google Cloud, cloud.google.com/learn/what-is-artificial-intelligence.
- Grace, Katja, et al. "Thousands of AI Authors on the Future of AI." arXiv, 2024, arxiv.org/abs/2401.02843.
- McKinsey & Company. (2023, June 14). *Economic potential of generative AI* | McKinsey. [www.mckinsey.com](https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier); McKinsey. <https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier>
- Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023). The Impact of AI on Developer Productivity: Evidence from GitHub Copilot. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2302.06590>
- Stack Overflow. "2025 Developer Survey." Stack Overflow, 2025, survey.stackoverflow.co/2025.
- Stanford Human-Centered Artificial Intelligence. "AI Index Report 2026." Stanford HAI, 2026, aiindex.stanford.edu.

The Future of Jobs Report 2025. (2025, August 25). World Economic Forum.

<https://www.weforum.org/publications/the-future-of-jobs-report-2025>

U.S. BUREAU OF LABOR STATISTICS. (2025, April 18). Software Developers, Quality Assurance Analysts, and Testers. Bls.gov; U.S. Bureau of Labor Statistics.

<https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm>

World Economic Forum. (n.d.). "A Brief History of Artificial Intelligence." World Economic Forum, [weforum.org](https://www.weforum.org).